DATA ANALYTICS WITH POWER BI PROJECT REPORT

(Project Semester August-December 2025)

SUSTAINABLE PRODUCT BENCHMARK DASHBOARD

Submitted by

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CERTIFICATE

This is to certify that Gokul G bearing Registration no. 12306197 has completed INT374 project titled, **"SUSTAINABLE PRODUCT BENCHMARK DASHBOARD"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of …….

Lovely Professional University

Phagwara, Punjab.

Date:

DECLARATION

I, GOKUL G , a student of Bachelor of Technology under CSE discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own work and is genuine

GOKUL G

Registration Number: 12306197

Introduction

With increasing awareness about environmental sustainability, organizations are required to evaluate their products not only on performance and market demand but also on their environmental impact. Data-driven analysis helps in understanding how different factors such as material type, manufacturing process, and regional demand influence sustainability outcomes.

This project focuses on analyzing sustainable product performance using Power BI. An interactive dashboard has been designed to evaluate products across multiple dimensions such as carbon footprint, water usage, recyclability, durability, consumer satisfaction, and market demand. The dashboard enables users to explore sustainability metrics in a simple and visually intuitive manner.

Project Objectives

The primary objectives of this project are:

- To analyze sustainability performance across product categories and regions
- To measure environmental impact using carbon footprint, water usage, and energy consumption
- To compare durability and recyclability across different material types
- To evaluate consumer satisfaction in relation to pricing and energy usage
- To identify optimum and non-optimum products using defined sustainability criteria
- To build an interactive Power BI dashboard for easy decision-making

Dataset Overview

The dataset used in this project is a comprehensive sustainability-focused product dataset.

Dataset Summary

- **Number of Records:** 52,425
- **Number of Features:** 25
- **File Format:** CSV
- **Tool Used:** Power BI

The dataset captures product-level information along with environmental, consumer, market, and regulatory metrics.

Tools and Technologies Used

- **Power BI Desktop** – Data modeling and dashboard creation
- **CSV Dataset** – Data source
- **DAX** – Used for calculated measures such as averages, optimum rate, and aggregations

This project was developed in an academic environment under the guidance of a Power BI instructor and follows a student-friendly approach.

Dashboard Design Explanation

Figure 1: Sustainable Product Benchmark Dashboard – Overview

The *Sustainable Product Benchmark Dashboard* is designed using a **dark theme** to enhance readability and visual contrast. The layout follows a logical top-to-bottom flow starting with KPIs, followed by category-wise, material-wise, regional, and consumer analysis.

Dashboard Components Analysis

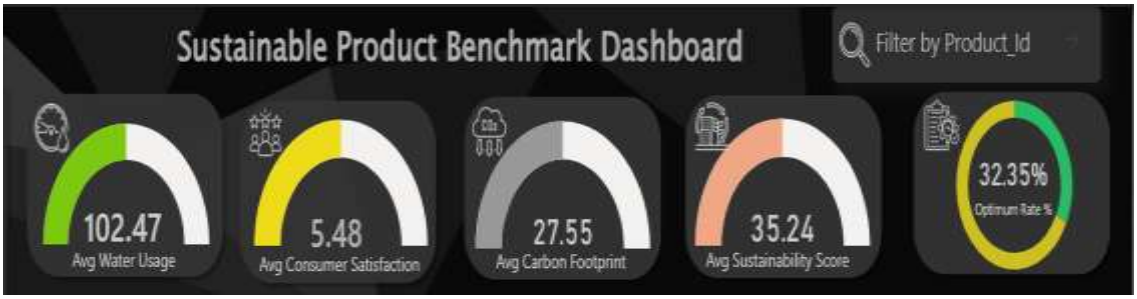
- **Interactive Filters**

The dashboard includes the following slicers:

- Category
- Material Type
- Region
- Manufacturing Process
- Price Range (Slider)
- Is Optimum (Toggle)
- Product ID Search

These filters allow users to dynamically explore specific products or segments.

- **Key Performance Indicators (KPIs)**



The top section of the dashboard displays key sustainability indicators:

- **Average Water Usage:** 102.47
- **Average Consumer Satisfaction:** 5.48
- **Average Carbon Footprint:** 27.55
- **Average Sustainability Score:** 35.24
- **Optimum Rate:** 32.35%

These KPIs provide a quick snapshot of overall product sustainability performance.

➢ **Carbon Footprint by Category**

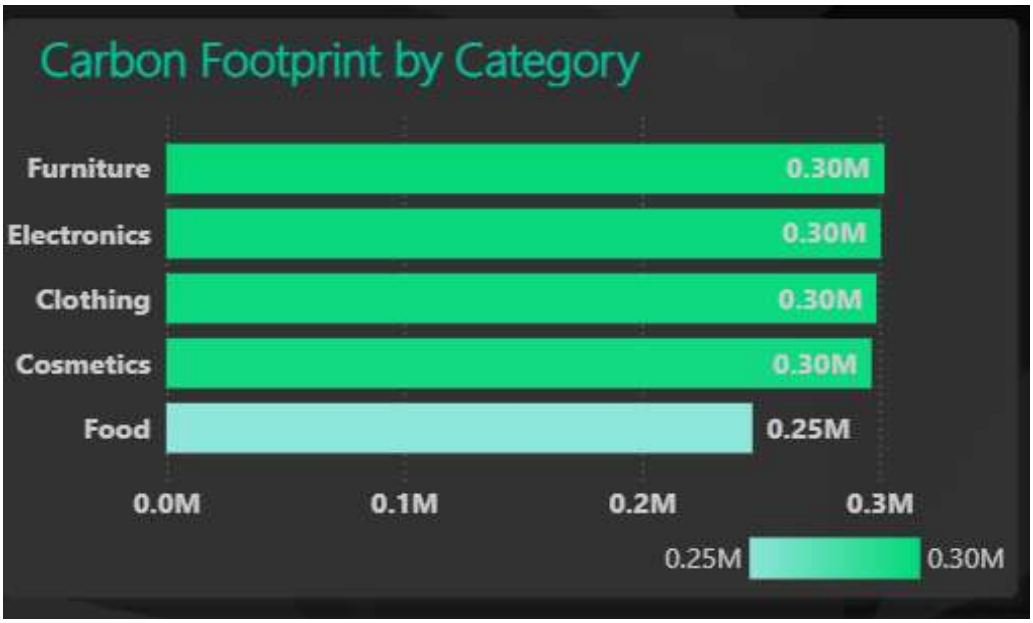


Figure 3: Carbon Footprint Comparison Across Product Categories

This bar chart compares carbon footprint across product categories:

- Furniture
- Electronics
- Clothing
- Cosmetics
- Food

The visualization helps identify categories contributing more to carbon emissions, enabling targeted sustainability improvements.

- **Durability and Recyclability by Material Type**

  This section includes:

  - A scatter plot showing durability across material types
  - A donut chart representing material distribution

  The analysis highlights how organic and recyclable materials generally perform better in sustainability-related metrics compared to synthetic materials.

- **Region-Wise Carbon Footprint**

Figure 4: Region-wise Carbon Footprint Distribution

A filled map visual represents carbon footprint distribution across regions such as:

- China
- EU
- India
- US

This allows geographic comparison of environmental impact and helps identify high-impact regions.

- **Market Demand by Price**

A donut chart divides market demand into:

- Low
- Medium
- High

The visualization shows a nearly balanced distribution, indicating that price alone does not dominate demand and sustainability may influence purchasing decisions.

- **Consumer Satisfaction Analysis**

A scatter plot analyzes the relationship between:

- Energy Consumption
- Price

This visualization helps understand how operational efficiency and pricing affect consumer satisfaction levels.

Key Insights

- Certain product categories have consistently higher carbon footprints
- Organic and recycled materials show better sustainability performance
- Only about one-third of products qualify as optimum, indicating improvement potential
- Higher price does not always guarantee higher consumer satisfaction
- Regional variations significantly impact sustainability metrics

Conclusion

This project demonstrates how Power BI can be effectively used to analyze and visualize sustainability-related data. The dashboard provides a holistic view of product performance by combining environmental impact, consumer behavior, and market demand. It enables stakeholders to identify improvement areas and supports data-driven sustainability decisions.

Future Enhancements

- Integration of real-time sustainability data
- Predictive analysis for future market demand
- Advanced optimization recommendations using machine learning
- Expansion of dataset with additional regions and product categories